



M. Paola Vaccaro

Born in 1997 in Larino (CB), Italy
Currently in Heidelberg, Germany

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Interests & Expertise

- Main interests:
 - Computational astrophysics
 - Gravitational-wave astronomy
 - Black hole dynamics
 - Stellar clusters
 - Stellar evolution
- Data analysis and visualization
- Scientific software development
- Methods:
 - Semi-analytical models
 - Hydrodynamics
 - Population synthesis
 - Cosmological simulations analysis
 - Bayesian statistics
- Numerical tools:
 - Python*
 - C/C++
 - Fortran
 - Linux
 - MPI
 - CUDA
 - Slurm*
 - HPC workflows*(*** = proficient)
- Scientific writing in LaTeX
- Conference presentations
- Science communication and outreach
- Collaborative research
- Mentoring
- Project management
- Languages:
 - Italian (mother tongue)
 - English (fluent, level C1)
 - Spanish (basic, level A2)
 - German (basic, level A1)

Education

Sep. 2023 – 2027 PhD in Astronomy and Cosmic Physics

At Institute for Theoretical Astrophysics (ITA), ZAH,
Heidelberg University in Heidelberg, Germany.

Supervisor: Prof. Michela Mapelli - Thesis title: “Binary Black Hole population in the Local Universe”

Oct. 2020 – Sep. 2022 Master degree in Astrophysics

At Università degli Studi di Padova in Padua, Italy.

Final grade: 110/110 *cum laude*, GPA: 29.4/30

Supervisor: Prof. Michela Mapelli - Thesis title: “Hierarchical Black Holes Mergers in Active Galactic Nuclei”

Sep. 2019 – Jul. 2020 Master courses in Matter Physics

At Università di Pisa in Pisa, Italy - GPA: 28.8/30

Sep. 2016 – Sep. 2019 Bachelor degree in Physics

At Università di Pisa in Pisa, Italy.

Final grade: 110/110 *cum laude*, GPA: 28.6/30

Supervisor: Prof. Pier Giorgio Prada Moroni - Thesis title: “Cooling evolution of white dwarfs and cosmochronology”

Sep. 2011 – Jul. 2016 High school diploma

At Liceo Scientifico Alfano da Termoli in Termoli (CB), Italy.

Final grade: 100/100 *cum laude*

Aug. 2014 – Jul. 2015 High school exchange year

At Mead High School in Mead, Colorado, USA.

GPA: 4.0/4 - ACT score: 33/36

Scientific publications *as main author*

- (IV) **AGN-driven BBH mergers: Black hole populations and hierarchical growth across the AGN parameter space**, [arxiv:2606.10823](#)
- (III) **The role of migration traps in the formation of binary black holes in AGN disks**, [A&A, 708, A171 \(2026\)](#)
- (II) **Astrophysics: a modern discipline with a Newtonian origin**, [Phil. Trans. R. Soc. A, 383: 20230294. \(2025\)](#); Invited contribution to a theme issue.
- (I) **Impact of gas hardening on hierarchical black hole mergers in AGN disks**, [A&A, 685, A51 \(2024\)](#)

Scientific publications

as co-author

- **The Science of the Einstein Telescope**, Adrian Abac et al. (incl. MPV), 2025, arXiv:2503.12263
- **The boring history of Gaia BH3 from isolated binary evolution**, Giuliano Iorio et al. (incl. MPV), A&A, 690, A144 (2024)
- **Hierarchical binary black hole mergers in globular clusters: mass function and evolution with redshift**, Stefano Torniamenti et al. (incl. MPV), A&A, 688, A148 (2024)
- **GW190521: a binary black hole merger inside an active galactic nucleus?**, Sophia Morton et al. (incl. MPV) 2023, Phys. Rev. D, 108, 123039
- **The hitchhiker's guide to the galaxy catalog approach for gravitational wave cosmology**, Jonathan R. Gair et al. (incl. MPV), 2022, AJ, 166, 22

Funding and affiliations

Member of IMPRS ([October 1st, 2023 – Present](#)) PhD fellow of the 19th generation in the International Max Planck Research School (IMPRS) for Astronomy and Cosmic Physics at Heidelberg University.

Member of the Einstein Telescope collaboration ([October 1st, 2023 – Present](#)), Division 3

Member of the Young Research Convent ([October 1st, 2023 – Present](#)) of the Heidelberg Cluster of Excellence STRUCTURES. Program funded by the German Excellence Strategy fund.

Travel grant ([April 2026](#)) Awarded through the ACME (Astrophysics Centre for Multimessenger studies in Europe) Transnational Access programme for a research visit to the GSSI in L'Aquila, Italy.

Travel grant ([July 2025](#)) Participation in the IAU Symposium 398 and MODEST-25 hosted by the Seoul National University in Seoul, South Korea, sponsored by the International Astronomical Union.

Travel grant ([July 2024](#)) Participation in the International HPC Summer School hosted by the RIKEN Center in Kobe, Japan, sponsored by the EuroHPC Joint Undertaking via the HPC SPECTRA project.

Pre-doctoral fellowship ([February 1st – August 31st, 2023](#)) in DEMOBLACK ERC Team at Università degli Studi di Padova, Padua, Italy. Supervisor: Prof. Michela Mapelli.

Internship in GW group ([October 1st – December 31st, 2022](#)) at IFAE-UAB, Barcelona, Spain. Supervisors: Dr. Lluïsa-Maria Mir i Martínez and Prof. Mario Martínez Pérez

Teaching experience and academic service

- Reviewer for *Astronomy & Astrophysics (A&A)* since 2025
- Reviewer for *Journal of Cosmology and Astroparticle Physics (JCAP)* since 2025
- Reviewer for *Physical Review D (PRD)* since 2026
- Teaching assistant for “Introduction to Astronomy and Astrophysics”: Summer Semesters 2024, 2025
Conducted daily exercise tutorials, corrected student submissions, and provided detailed feedback.
- Student co-supervision
 - Master student Julian L. Saling, “Hydrodynamical simulations of massive-star collisions” 2025/2026
 - Bachelor student Yannick Seif, “Pair Up of Binary Black Holes in Active Galactic Nuclei” 2024/2025
 - Bachelor student Jona A. Schubert, “The mass ratio of binary black holes: impact of mass transfer” 2024
- Volunteer staff member at EAS Annual Meetings: July 2024, July 2026
Supported registration and session logistics, including presentation slide checks and room coordination.

Conferences and workshops

- [Poster Presentation](#) | **EAS annual meeting** SwissTech Convention Centre, Lausanne, Switzerland
Session: “SS31: Exotic populations in star clusters seen through the lens of binary interactions”. Poster title: “Multi-Stage Collision Products in Dense Clusters”. 29 June–3 July, 2026
- [Contributed Talk](#) | **XIV Einstein Telescope Symposium** RWTH Aachen, Germany
Talk title: “Cosmic Evolution of Binary Black Hole Mergers from AGN Disks”. June 15th–19th, 2026
- [Contributed Talk](#) | **MODEST-25 & IAU Symposium 398** Seoul National University, Seoul, South Korea
Talk title: “Hierarchical Black Hole Mergers in AGN Disks”. June 16th–20th, 2025
- [Contributed Talk](#) | **Dynamix** Institute of Astronomy, Cambridge, UK
Title: “Hierarchical BH Mergers in AGN Disks: Tracing MBH Growth Across Cosmic Time”. June 2nd–6th, 2025
- [Poster Presentation](#) | **MODEST-24** Warsaw, Poland
(Remote attendance) August 19th–23rd, 2024
- [Invited Talk](#) | **New ideas on the origin of black hole mergers** Niels Bohr Institute, Copenhagen, Denmark
Talk title: “Specks of light from AGNs as possible indicators of the dynamical formation of binary black holes”. August 12th–16th, 2024
- [Contributed Talk](#) | **EAS annual meeting** Congress Center, Padova, Italy
Session: “S1: Unveiling black hole Growth across Cosmic Time in the JWST and LISA era”. July 1st–5th, 2024
Talk title: “Hierarchical Black Hole Mergers in Active Galactic Nuclei Discs”.
- [Poster Presentation](#) | **EAS annual meeting** Congress Center, Padova, Italy
Session: “SS2: Spacetime ripples and light from compact binary mergers: Lessons learned from O4 so far, and a look at the future”. Poster title: “Specks of light from AGNs as possible indicators of dynamical formation of BBHs”. July 1st–5th, 2024
- [Contributed Talk](#) | **Massive Black Holes in the First Billion Years** University College Cork, Cork, Ireland
Talk title: “Possible constraints on the SMBH mass distribution from hierarchical BBH mergers in AGN discs”. April 29th – May 3rd, 2024
- [Contributed Talk](#) | **RESCEU-NBIA workshop on GW sources** University of Tokyo, Tokyo, Japan
Talk title: “The impact of Gas Hardening on Hierarchical Black Hole Mergers in Active Galactic Nuclei Disks”. Contribution to the [discussion panel](#) “Many-Body Systems”. December 7th–15th, 2023
- [Flash Talk](#) | **Workshop "GwPopNext"** University of Milano Bicocca, Milan, Italy
Talk title: “Semi-analytical Models for Hierarchical Binary Black Hole Mergers”. July 10th–14th, 2023
- [Poster Presentation](#) | **Renaissance of Stellar Black-Hole Detections** Lorentz Center, Leiden, Netherlands
Invited participation June 26th–30th, 2023
- [Invited Talk](#) | **Physics of Data Workshop** Venetian institute of science, literature and art, Venice, Italy
Talk title: “How do black holes grow? Active galactic nuclei as a breeding ground for intermediate-mass black holes” April 19th–21st, 2023

Colloquia

- Department Colloquium GSSI, L'Aquila, Italy | April 24th, 2026
- Institute Seminar ISTA, Vienna, Austria | November 13th, 2025
- “BinBoom” Group Meeting (online) The University of Arizona | September 30th, 2025
- Physics of Stellar Objects Group Meeting HITS, Heidelberg, Germany | April 16th, 2025
- Gravitation and Cosmology Group Meeting GSSI, L'Aquila, Italy | December 9th, 2024
- ITP Cosmology Seminar ITP, Heidelberg, Germany | October 29th, 2024
- Einstein Telescope monthly meeting (online) ET-OSB Div3 | October 25th, 2024

- “Astroscopic” Seminar (online) GSSI, L’Aquila, Italy | April 26th, 2024
- Galaxy Group Meeting ITA-ZAH, Heidelberg, Germany | March 27th, 2024
- Planet Formation Group Meeting ITA-ZAH, Heidelberg, Germany | February 20th, 2024

Schools and workshops attendance

- “AI for Astronomy” Summer School hosted by the IWR in Heidelberg, Germany September 8th-12th, 2025
- International HPC Summer School hosted by the RIKEN Center in Kobe, Japan July 7th-12th, 2024
- “ML4ASTRO, Machine Learning in Astrophysics” workshop hosted by HITS in Heidelberg June 2024; July 2025
- “360° approach to Common Envelope Evolution: From Binary Progenitors to Remnants” hosted by Institut de Ciències del Cosmos of the Universitat de Barcelona (remote attendance) June 10th-14th, 2024
- 4th G2Net Training School “A network for Gravitational Waves, Geophysics and Machine Learning” hosted by Aristotle University of Thessaloniki (remote attendance) March 28th-30th, 2023
- Python Course (level advanced) hosted by ICFO in Barcelona, Spain November 24th-30th, 2022
- GW Open Data Workshop hosted by GWOSC (online event) May 23rd-25th, 2022
- IPAM Workshop III: Source inference and parameter estimation in Gravitational Wave Astronomy (remote attendance) November 15th-19th, 2021

Outreach and Media Coverage

- Research [featured in Gist.Science](#): “The formation of binary black holes in AGN disks”. April 8th, 2026
- Member of [Astronomers for Planet Earth](#) since 2025
- Author of “The Origins of Black Hole Mergers: Diverse Pathways Across the Universe”, March 7th, 2025
[blog post](#) for the STRUCTURES Cluster of Excellence, aimed to a general audience.
- Co-organizer for the [workshop](#) “A dance of stars and black holes” of the Girl’s Day program 2024, 2025, 2026
“Physik hautnah!” for school-aged students
- Co-author (with Dr. Giuliano Iorio and Dr. Erika Korb) of a pamphlet about stellar evolution, April 2024
distributed to 12th-grade high schoolers in the school Liceo Volta (Reggio Calabria, Italy) by Scholé
- Speaker and staff member at the 2023 Scholé Summer School titled “PHYSIS”. Talks:
 - “Understanding Nature with Quantum Mechanics” with Prof. Giancarlo Cella (Pisa, Italy) July 25th, 2023
 - “The concept of space from Newton to Einstein” with Prof. Paolo Bussotti (Udine, Italy) July 29th, 2023
- Member of the organizing committee for the Scholé “School of Physics” for high schoolers since 2022
- Staff member of the cultural association [Scholé](#) based in Roccella Jonica (RC), Italy since 2020